Part 4. Selection of Charts and Encoding

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Charts for each question

Each chart answers one of Natalia's questions and was chosen by its function in the Visual Vocabulary. Every chart passes the DRACO hard rules that apply to it; the map is outside DRACO and is judged with the course's GeoHandout.

Question	Function	Chart
Q1 busiest months	Evolution over time	C1 line chart of guests per month, July 2015 to August 2017, one line per hotel
Q5 cancellations	Evolution over time	C2 line chart of the cancellation rate per month
Q7 price	Evolution over time	C3 line chart of the median price per night per month
Q2 countries	Ranking, spatial	C4 sorted horizontal bars, top 10 countries and Other, one panel per hotel; C5 small map
Q3 season per market	Evolution over time	C6 line chart of the selected country's guests per month against all guests
Q4 length of stay	Magnitude	C7 sorted bars of median nights per country
Q6 booking advance	Magnitude	C8 sorted bars of median lead time per country

C1 to C3 use the full timeline instead of pooled calendar months, because only 2016 is complete; C6 pools months, but it compares a country with all guests over the same months, so the extra July and August weigh on both lines alike. Bars start at zero and are sorted, so position and length, the most accurate channels, carry every value, and medians are used because the data is skewed.

Encoding, colour and accessibility

The two hotels use blue and purple from Paul Tol's bright palette (Figure 3). Both pass the 3:1 contrast for graphics, but with Natalia's deuteranopia they are almost equally light, so each hotel also has its own line style, point shape and a direct label at the end of its line: redundant coding (Ware G5.11) and Gestalt proximity replace the legend. Bars use a single dark blue, so no fill patterns are needed. C2 and C3 repeat the time axis of C1 directly below it, so a month is read in the same place in the three charts (Gestalt proximity and similarity), and the country panels share one scale, so the hotels are compared by position. All 28 items of our accessibility plan were reviewed: twelve are solved by this design and the rest are planned for Parts 5 and 6.

Prototype

Both screens share one frame (Figures 1 and 2): title, skip and help links, four KPIs, a filter panel on the right with the active filters and a reset button, and a footer with

the source and the duplicates note. The main screen answers Q1, Q5 and Q7 with C1 as its focal point, taking 46 % of the content area. The detail screen answers Q2 and Q3 with C4 and C6; Q4, Q6 and the map wait behind “Show more”, so no screen shows more than three charts, following the cognitive accessibility limits.

Figures

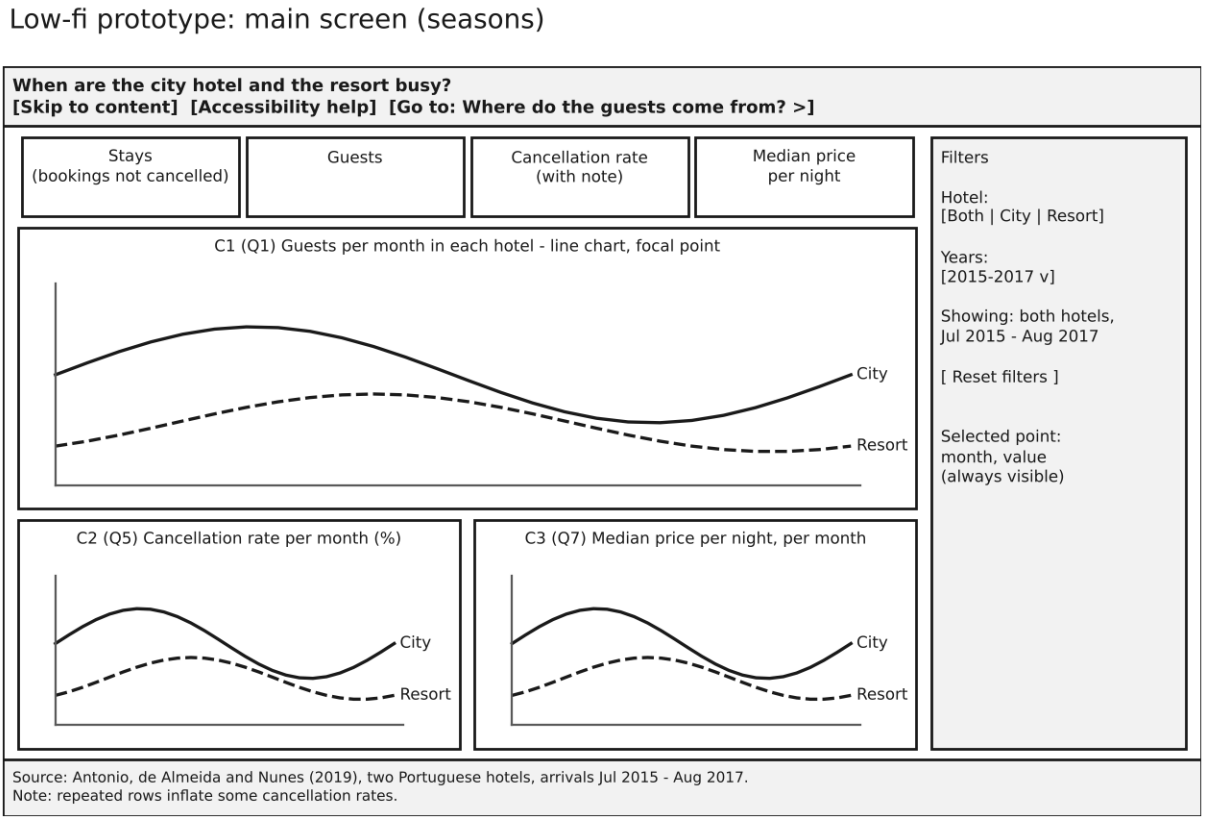


Figure 1. Low-fidelity prototype of the main screen, 1280 × 800 px. Source: our design; code: lowfi_prototype.ipynb.

Low-fi prototype: detail screen (source markets)

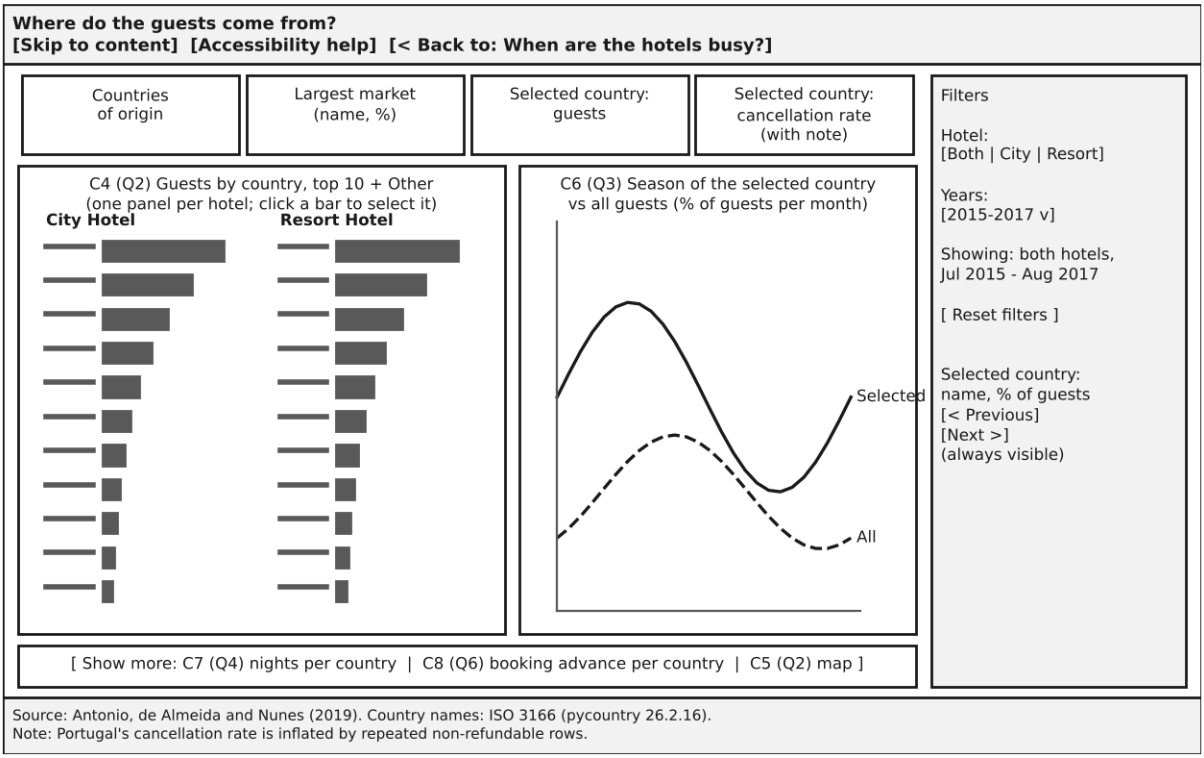


Figure 2. Low-fidelity prototype of the detail screen. Source: as Figure 1.

Dashboard colours with normal vision and with deuteranopia

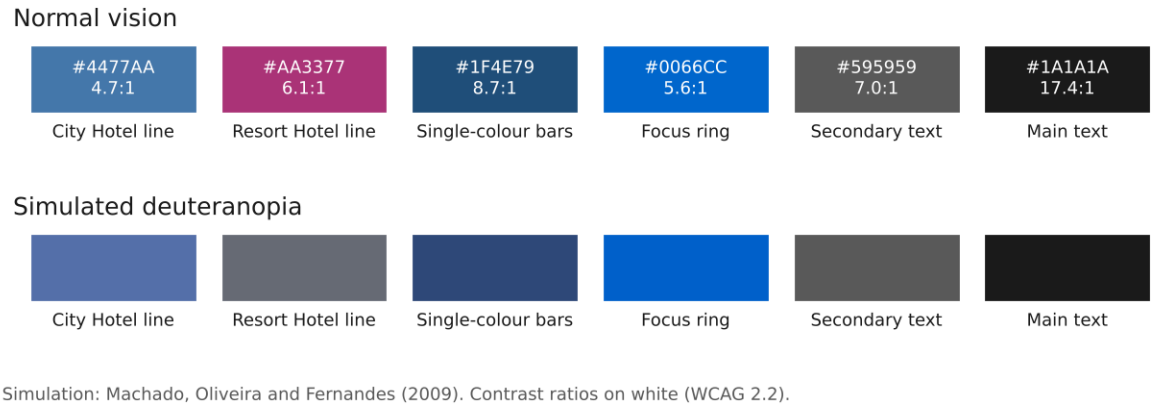


Figure 3. Dashboard colours with normal vision and simulated deuteranopia. Source: encoding_checks.ipynb.